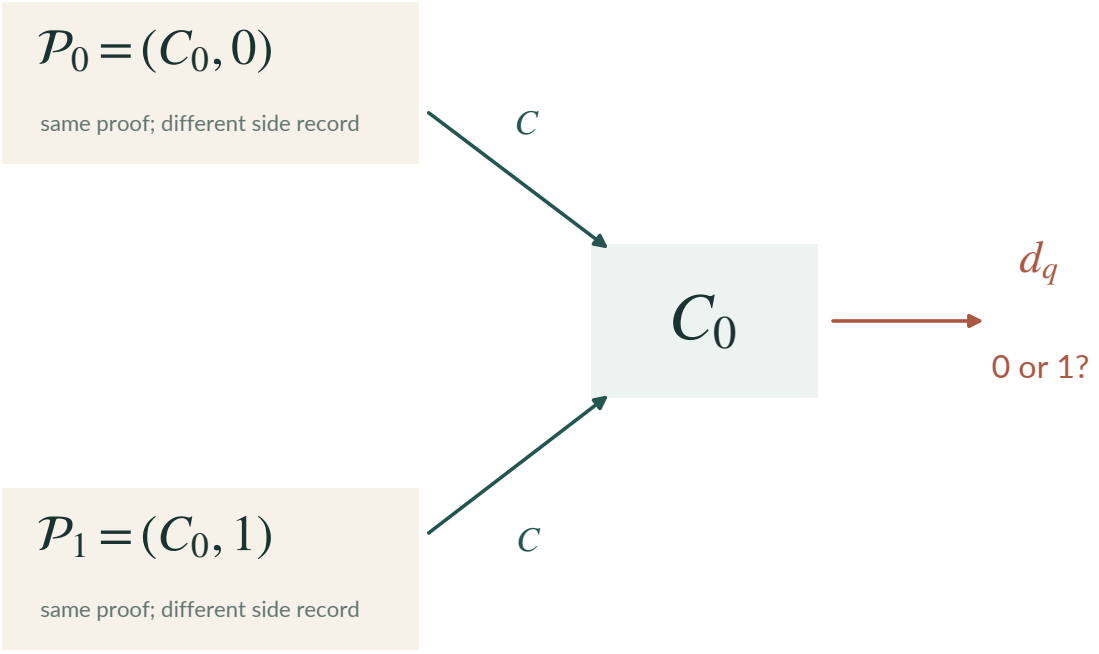

$$q(\mathcal{P}_0) = 0 \neq 1 = q(\mathcal{P}_1)$$

The decoder sees the same input. It cannot recover both different answers.